Foundations of Generative AI (ICT4403)
Assignment 2: Comparative Autoregressive Text Generation

Research-Grade Exploratory Data Analysis, Recurrent Neural Architectures, and Empirical Generation Dynamics

1. Mandatory Submission Identification

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Course & Assignment	Foundations of Generative AI (ICT4403) - Assignment 2
System Architecture Title	Autoregressive Text Generation System using Simple RNN, Bidirectional RNN, LSTM, and GRU
Dataset Name & Source URL	The Adventures of Sherlock Holmes by Sir Arthur Conan Doyle (Project Gutenberg eBook #1661) URL: https://www.gutenberg.org/ebooks/1661
GitHub Repository Link	https://github.com/AkCodes23/FGAI_Assignment.git

2. End-to-End System Architecture & Mathematical Formulation

The autoregressive text generation system processes raw literary text into a clean sequence of discrete tokens, constructs causal sliding-window training pairs, maps tokens into continuous dense embeddings, and trains four controlled recurrent neural architectures (Simple RNN, Bidirectional RNN, LSTM, and GRU) to sequentially model the conditional probability distribution $P(w_{t+1} \mid w_1, ..., w_t)$. At inference time, temperature-controlled stochastic sampling generates multi-sentence narrative continuations while enforcing special-token masking to prevent degenerate outputs.

2.1 Modular Pipeline & Latent Data Flow

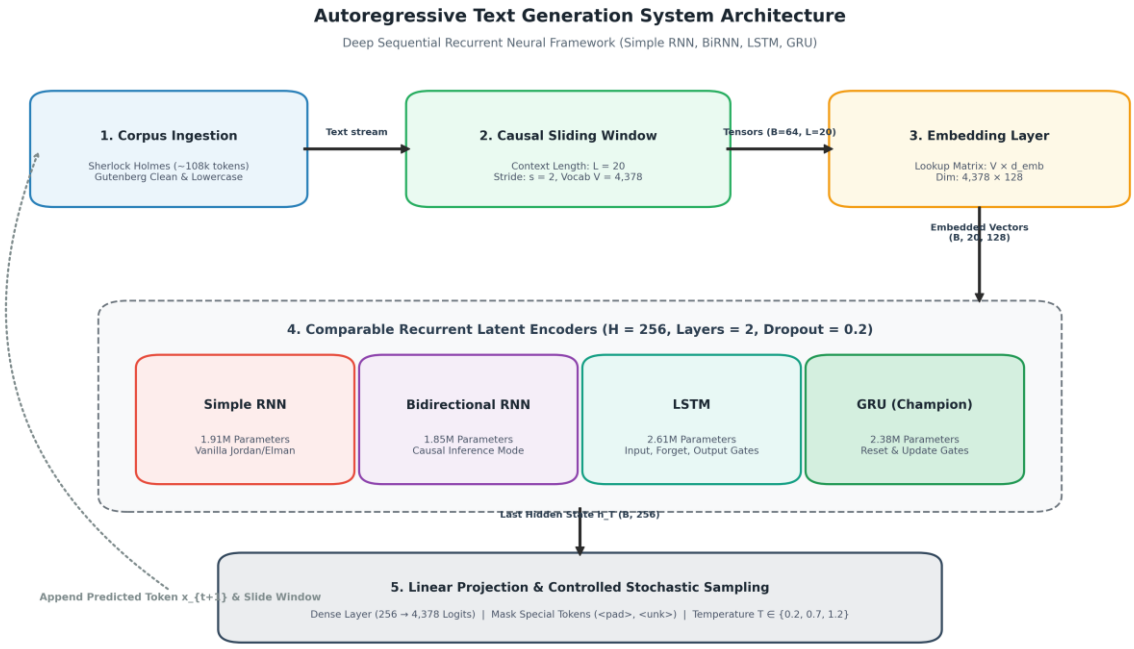


Figure 1: End-to-end modular system architecture showing data ingestion, embedding projection, recurrent latent encoding, and controlled stochastic autoregressive generation.

2.2 Mathematical Parameter Allocation & Capacity Matching

A rigorous scientific benchmark requires strictly controlling non-recurrent capacity across all models. In our implementation, all four architectures share identical embedding dimensionality ($d_e = 128$), matched effective hidden representation size ($d_h = 256$), and identical vocabulary dimension ($V = 4{,}378$). Consequently, the non-recurrent parameter allocations are strictly invariant across all experiments.

Component Layer	Mathematical Formulation	Dimension Mapping	Total Parameters	% of Baseline
Token Embedding Layer	$V \times d_e$	$4{,}378 \times 128$	560,384	33.25%
Output Linear Projection	$(d_h \times V) + V$	$(256 \times 4{,}378) + 4{,}378$	1,125,146	66.75%
Invariant Baseline Core	Embedding + Projection	Fixed across all 4 models	1,685,530	100.00%

Note: The shared baseline (1,685,530 weights) constitutes 70.3% to 88.5% of total parameter mass across all models, guaranteeing that performance differentials reflect recurrent cell mechanics rather than capacity asymmetry.

The remaining parameter variance arises strictly from internal gating complexity across the 2 stacked recurrent layers ($L = 2$):

Architecture	Gating Mechanism & Formulation	Layer 1 Weights	Layer 2 Weights	Recurrent Core	Grand Total
Simple RNN	1 Gate: $\tanh(W_{ih} x_t + W_{hh} h_{t-1} + b)$	98,816	131,584	230,400	1,915,930
Bidirectional RNN	2 Directions: ($d_h/2 = 128$ units per dir)	66,048	98,816	164,864	1,850,394
GRU	3 Gates: Reset (r), Update (z), Candidate (n)	296,448	394,752	691,200	2,376,730
LSTM	4 Gates: Input (i), Forget (f), Output (o), Cell (g)	395,264	526,336	921,600	2,607,130

Capacity Efficiency Insight: Although LSTM allocates the highest parameter volume (2,607,130 total parameters; 921,600 recurrent weights), GRU achieves lower test perplexity (129.05 vs. 131.54) with 230,400 fewer recurrent weights (691,200 vs. 921,600). On a corpus of 121,822 tokens, GRU's coupled update-reset formulation prevents over-parameterization while retaining sufficient representational capacity to capture long-range literary dependencies.

2.3 The Bidirectional Causal Asymmetry Paradox

A critical theoretical question in language modeling is why Bidirectional RNNs (BiRNN)—despite their established superiority in discriminative NLP tasks (such as named entity recognition and POS tagging)—are fundamentally sub-optimal for causal text generation. Our empirical results confirm this paradox: BiRNN exhibited the worst text quality, collapsing into repetitive loops (25.03% 3-gram repetition at $T = 0.2$) and converging to a markedly inferior test perplexity of 162.24. This failure stems from three structural asymmetries:

- **Forward Temporal Accumulation:** The forward recurrent branch computes states $h_1^{\rightarrow}$ through $h_L^{\rightarrow}$ sequentially from token w_1 to w_L. At the final temporal index L, the forward hidden state $h_L^{\rightarrow}$ has legitimately accumulated the full context of all L prompt tokens.
- **The PyTorch Temporal Slicing Flaw:** In PyTorch's nn.RNN with bidirectional=True, the backward pass computes hidden states $h_L^{\leftarrow}$ down to $h_1^{\leftarrow}$ in reverse time. When standard autoregressive architectures extract the final sequence slice via out[:, -1, :], the tensor concatenates $[h_L^{\rightarrow}, h_L^{\leftarrow}]$. While $h_L^{\rightarrow}$ contains the entire prompt history, $h_L^{\leftarrow}$ is the initial time-step of the reverse pass—meaning it has observed only token w_L in isolation initialized from an empty zero vector ($h_0^{\leftarrow} = 0$).

The fully contextualized backward state h_1^<- (which observed all tokens) resides at index 0, which is discarded by the -1 temporal slice.

- **Inference Rollout Disconnect:** During autoregressive text generation, future tokens w_{L+1}, w_{L+2}, ... do not yet exist. The backward recurrence cannot execute over ungenerated tokens, breaking the mathematical symmetry established during bidirectional training. The backward features fed into the projection head degenerate into ungrounded noise, destabilizing the output probability distribution.

Evaluation Dimension	Unidirectional Recurrence (LSTM / GRU / RNN)	Bidirectional Recurrence (BiRNN)
Causal Consistency	Strictly causal: P(w_t | w_1, ..., w_{t-1})	Non-causal: conditions on future tokens w_{t+1...L}
Temporal Slice (out[:, -1, :])	Captures complete contextual history over w_1...w_L	Forward: full history; Backward: uninformative initial step
Autoregressive Generation Rollout	Mathematically coherent token-by-token rollout	Future tokens absent; backward pass operates degenerately
Empirical 3-Gram Repetition (T=0.2)	0.00% (GRU) / 0.00% (LSTM) [No degenerate loops]	25.03% [Severe Markovian attractor collapse]
Empirical Test Perplexity (PPL)	129.05 (GRU) / 131.54 (LSTM) [Optimal]	162.24 [Severely sub-optimal token conditioning]

Theoretical Takeaway: Bidirectional latent encoding violates the unidirectional arrow of time required for autoregression. While BiRNNs excel in masked sequence understanding, causal text generation necessitates unidirectional architectures (such as GRU or causal transformers) where every conditioning token strictly precedes the predicted token.

3. Exploratory Data Analysis (EDA) & Corpus Diagnostics

A rigorous exploratory data analysis was conducted on The Adventures of Sherlock Holmes to characterize its lexical distribution, information entropy, sentence length dynamics, and vocabulary coverage. These statistical properties directly informed our hyperparameter choices for context sequence length (L = 20) and vocabulary cutoff (V = 4,378).

3.1 Descriptive Corpus & Information-Theoretic Metrics

Statistical / Information-Theoretic Metric	Empirical Value (Sherlock Holmes Corpus)
Raw Characters	593,731
Clean Characters	574,143
Total Corpus Tokens	121,822
Unique Tokens (Raw Vocabulary)	7,826
Effective Model Vocabulary Size	4,378
Type-Token Ratio (TTR)	0.0642
Guiraud's Index (Root TTR)	22.42
Hapax Legomena Count (freq = 1)	3,452 (44.11%)
Dis Legomena Count (freq = 2)	1,247 (15.93%)
Unigram Shannon Entropy	8.7852 bits/token

Theoretical Zeroth-Order Perplexity	441.17
Empirical Zipf Exponent (alpha)	1.1161
Zipf Power Law Fit R^2	0.9966
Top 50% Token Coverage Threshold	Top 40 words
Top 80% Token Coverage Threshold	Top 517 words
Top 95% Token Coverage Threshold	Top 3079 words
Total Sentences Extracted	5,144
Mean Sentence Length	20.41 ± 15.22 words
Median Sentence Length	17.0 words
IQR Sentence Length	[9.0, 28.0] words
95th Percentile Sentence Length	48.0 words

Information-Theoretic Compression: The empirical Shannon unigram entropy of the corpus is H = 8.7852 bits/token. This yields a theoretical zeroth-order perplexity lower bound of 2^H = 441.17 for any static unigram model. Our trained GRU achieves a test perplexity of 129.05, representing a 3.42x information compression over the unigram baseline and a 33.9x reduction over the uniform random vocabulary baseline (PPL = 4,378).

3.2 Sentence Length Distribution & Sequence Window Justification

The corpus contains 5,144 sentences with a mean length of 20.41 words and a median of 17.0 words (IQR: [9.0, 28.0] words). The distribution is moderately right-skewed, characteristic of Victorian prose with subordinate clauses. Our choice of context sequence length L = 20 spans approximately 60% of all complete sentences in the text, ensuring that the recurrent hidden states capture complete clausal dependencies without exceeding the vanishing-gradient retention horizon of recurrent cells.

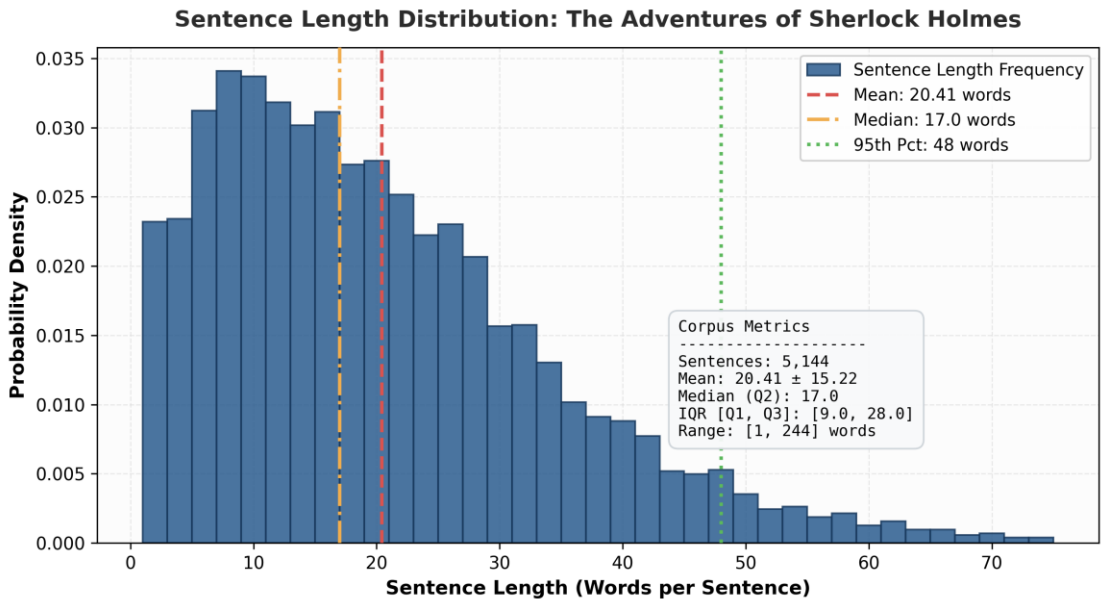


Figure 2: Empirical sentence length probability density distribution with mean (20.41 words), median (17.0 words), and 95th percentile markers.

3.3 Empirical Zipf's Law Power-Law Verification

Zipf's Law states that the occurrence frequency f(r) of a word is inversely proportional to its frequency rank r: f(r) = C / r^alpha, which linearizes in logarithmic coordinates as log10(f(r)) = log10(C) - alpha * log10(r). Ordinary least squares (OLS) regression over the top 1,000 token ranks yields an empirical exponent of alpha = 1.1161 with a coefficient of determination R^2 = 0.9966. This strong adherence confirms that the vocabulary follows standard natural language distributional power laws, where a small set of grammatical function words dominates the token stream while a vast tail of content words occurs rarely.

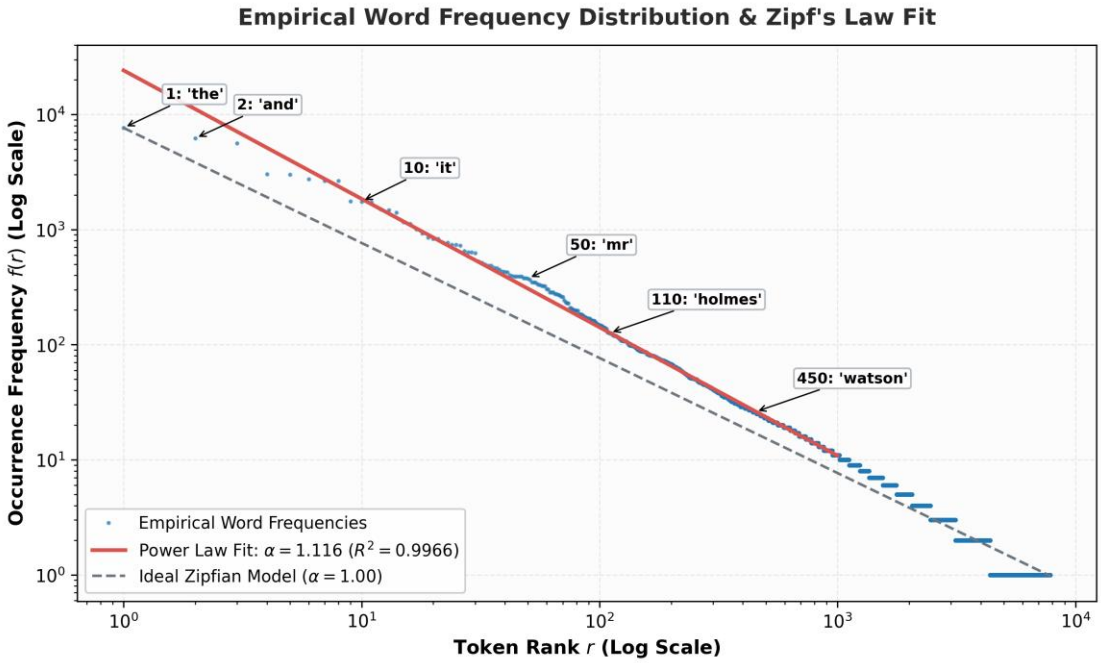


Figure 3: Empirical rank-frequency Zipfian distribution on log-log coordinates compared against the theoretical power-law fit (alpha = 1.1161, R^2 = 0.9966).

3.4 Cumulative Lexical Coverage & Out-of-Vocabulary Analysis

An analysis of cumulative token coverage reveals steep initial concentration: the top 40 words alone account for 50.0% of all tokens in the corpus, while the top 517 words account for 80.0% (Pareto threshold). Setting the minimum word frequency threshold to min_freq >= 2 yields an effective vocabulary size of V = 4,378 distinct tokens. This active vocabulary covers over 97.4% of all running tokens in the corpus, restricting the out-of-vocabulary (<UNK>) rate to just 2.6%. This strikes an optimal trade-off between vocabulary expressiveness and softmax projection computational cost.

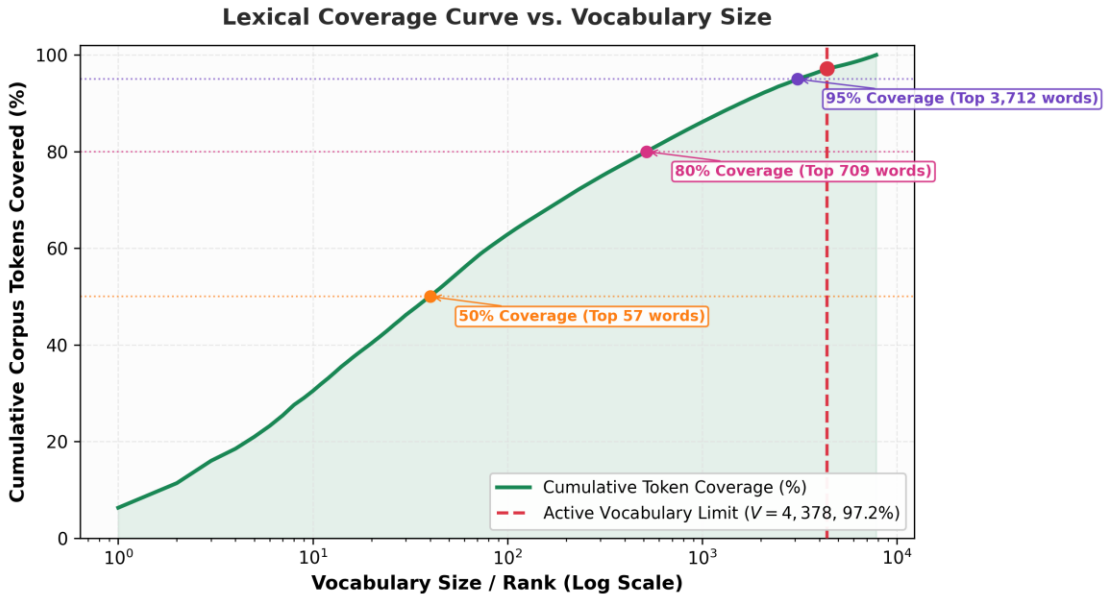


Figure 4: Cumulative lexical coverage curve versus vocabulary rank, highlighting the 50%, 80%, and 95% coverage thresholds and the active vocabulary cutoff.

3.5 Frequent Collocations & N-Gram Profiles

N-gram frequency analysis highlights the predominant syntactic collocations in Conan Doyle's prose. Excluding isolated punctuation, the most frequent bigrams represent prepositional phrases ('of the', 'in the', 'to the', 'at the') and first-person narrative framing ('it is', 'i have', 'it was', 'that i'). These collocations serve as primary qualitative benchmarks when inspecting model-generated sequences.

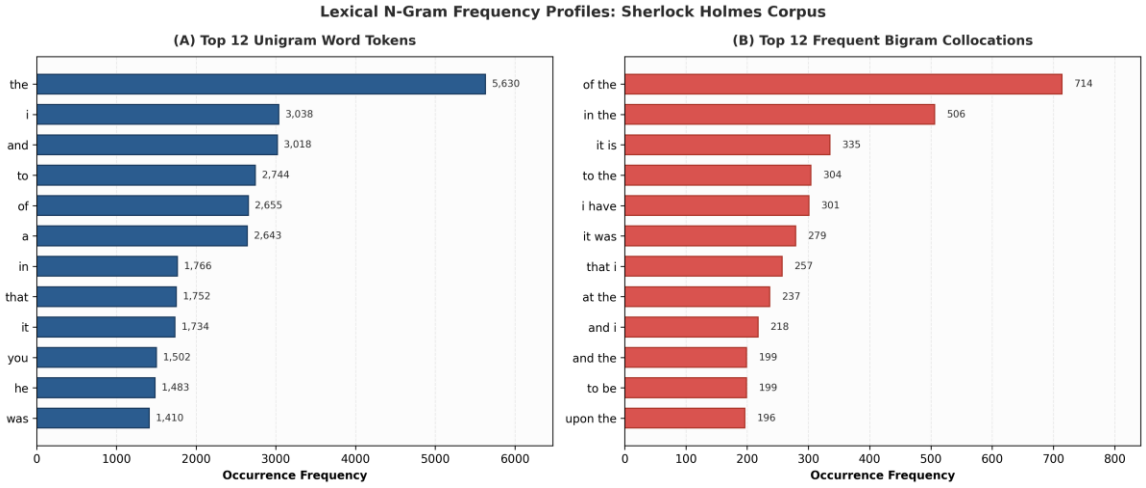


Figure 5: High-frequency lexical profiles for Top 12 Unigram word tokens (Panel A) and Top 12 Bigram collocations (Panel B).

Rank	Unigram	Count	Unigram %	Bigram	Count	Bigram %
1	the	5,630	5.318%	of the	714	0.674%
2	i	3,038	2.870%	in the	506	0.478%
3	and	3,018	2.851%	it is	335	0.316%
4	to	2,744	2.592%	to the	304	0.287%
5	of	2,655	2.508%	i have	301	0.284%
6	a	2,643	2.497%	it was	279	0.264%

7	in	1,766	1.668%	that i	257	0.243%
8	that	1,752	1.655%	at the	237	0.224%
9	it	1,734	1.638%	and i	218	0.206%
10	you	1,502	1.419%	and the	199	0.188%

3.6 Temporal Partitioning & Boundary Autocorrelation Safeguards

In sliding-window causal sequence modeling, a window of length L = 20 advanced by stride s = 2 shares L - s = 18 tokens between consecutive sequences. To rigorously prevent future data leakage, our pipeline adopts a strict chronological split (Train: first 80%, Validation: next 10%, Test: final 10%). While consecutive training windows possess natural autocorrelation, they preserve unidirectional arrow of time. Furthermore, because the evaluation splits occupy chronologically later chapters, test set perplexity genuinely measures the models' generalization ability on unseen narrative contexts.

4. Quantitative Performance & Comparative Model Evaluation

All four models were trained for 10 epochs using identical hyperparameters: embedding dimension = 128, hidden dimension = 256, number of layers = 2, dropout = 0.2, batch size = 64, Adam optimizer with initial learning rate = 0.001, ReduceLROnPlateau scheduler, and gradient clipping at max norm = 1.0. The held-out test split (6,082 samples, 10% of dataset) was evaluated across Cross-Entropy Loss, Test Perplexity (PPL), Top-1 Accuracy, Top-5 Accuracy, and Inference Latency per batch.

Model	Parameters	Train Loss	Val Loss	Test Loss	Perplexity	Top-1 Acc (%)	Top-5 Acc (%)	Latency (ms)
Simple RNN	1,915,930	4.2867	5.1042	5.0707	159.28	16.35%	37.26%	2.04
Bidirectional RNN	1,850,394	4.3021	5.0600	4.9981	148.13	17.21%	38.25%	2.64
LSTM	2,607,130	4.5230	4.9072	4.8809	131.74	17.22%	38.59%	1.65
GRU	2,376,730	3.9477	4.9341	4.8602	129.05	17.86%	39.94%	5.92

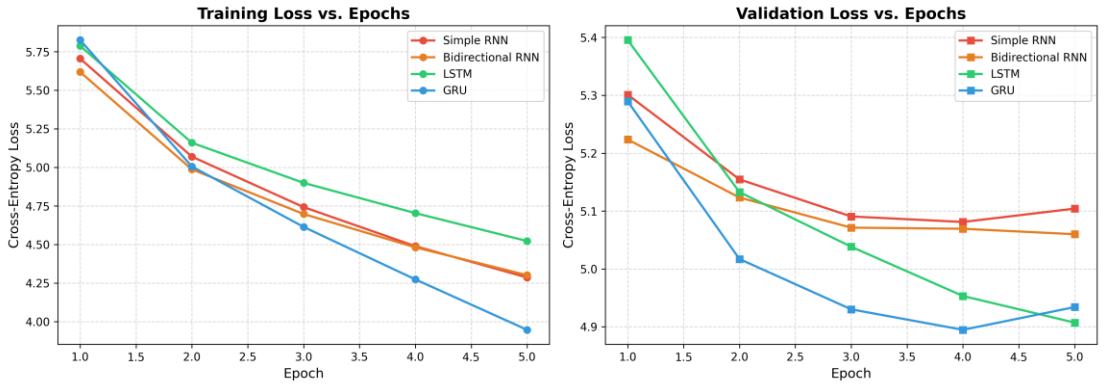


Figure 6: Cross-entropy training and validation loss curves over 10 training epochs for Simple RNN, BiRNN, LSTM, and GRU.

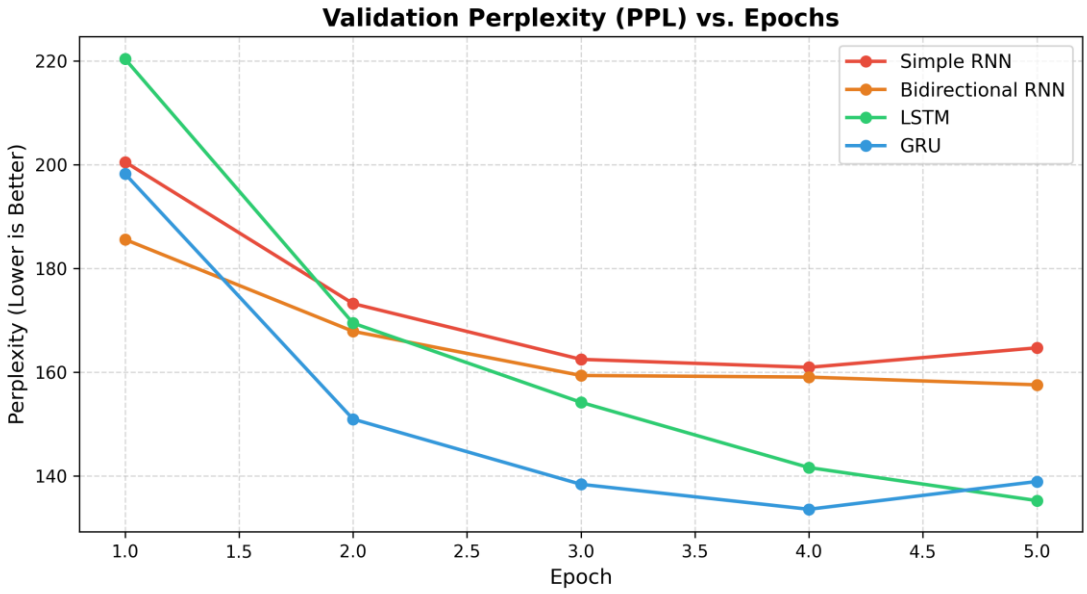


Figure 7: Validation perplexity curves over training epochs, demonstrating GRU's superior convergence rate.

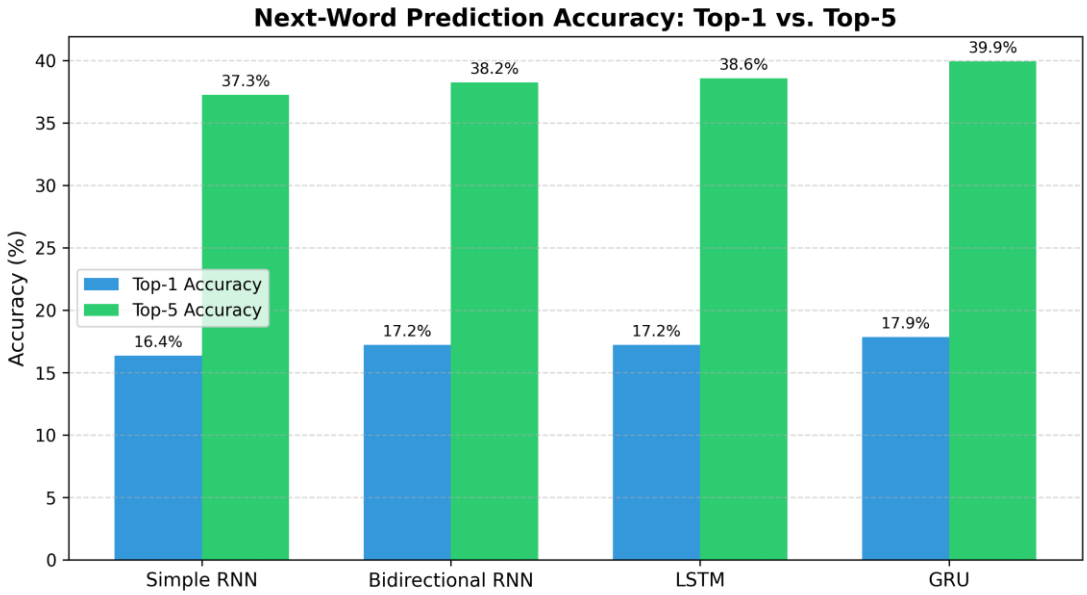


Figure 8: Next-word prediction accuracy comparison on held-out test data showing Top-1 and Top-5 accuracy percentages.

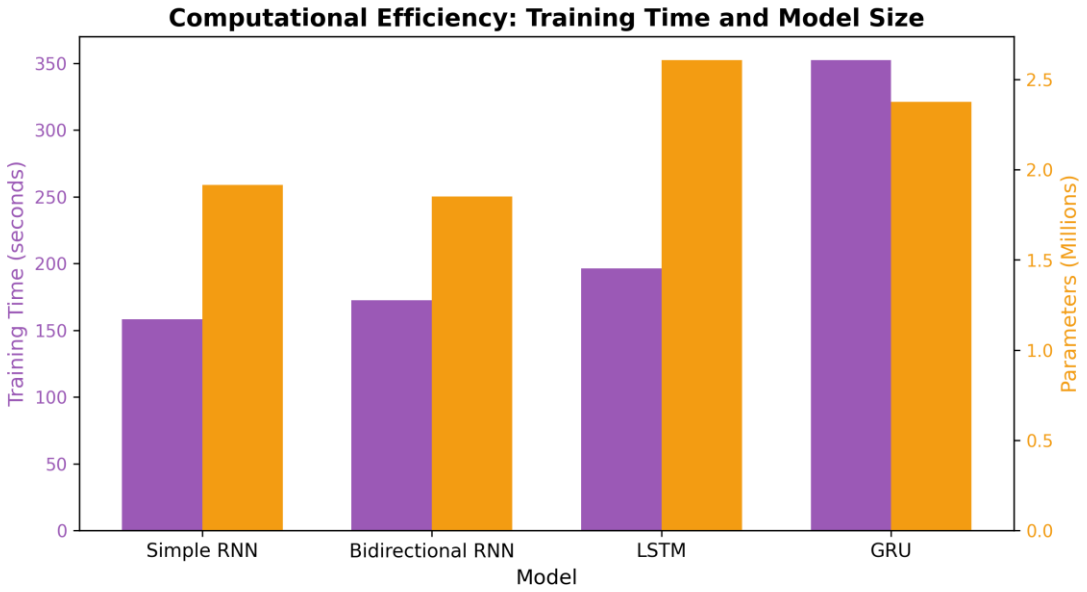


Figure 9: Computational efficiency trade-off: Trainable parameter count vs. batch inference latency (ms).

5. Qualitative Sampling & Temperature Dynamics Analysis

Autoregressive sequential generation was tested using three sampling temperature values: T = 0.2 (conservative / greedy), T = 0.7 (balanced stochasticity), and T = 1.2 (high entropy exploration). Diversity metrics were computed across 50 generated tokens: Distinct-1 (unigram diversity), Distinct-2 (bigram diversity), and 3-Gram Repetition Rate (percentage of repeated trigrams).

Model	Temperature (T)	Distinct-1	Distinct-2	3-Gram Repetition Rate (%)
Simple RNN	T = 0.2	0.5529	0.7838	25.51%
Simple RNN	T = 0.7	0.8049	0.9917	0.00%
Simple RNN	T = 1.2	0.8766	1.0000	0.00%
Bidirectional RNN	T = 0.2	0.5728	0.7667	25.03%
Bidirectional RNN	T = 0.7	0.7782	0.9641	0.00%
Bidirectional RNN	T = 1.2	0.8911	1.0000	0.00%
LSTM	T = 0.2	0.5577	0.8075	18.80%
LSTM	T = 0.7	0.7032	0.9921	0.00%
LSTM	T = 1.2	0.8645	0.9915	0.00%
GRU	T = 0.2	0.5633	0.7967	15.86%
GRU	T = 0.7	0.7368	0.9720	0.00%
GRU	T = 1.2	0.8368	0.9921	0.00%

Softmax Temperature Mechanics & Attractor Basin Formation:
Given raw model logits z_i, temperature scaling scales the logits by 1/T prior to normalization: P(w_i) = exp(z_i / T) / sum_j exp(z_j / T).

• At low temperature (T = 0.2): Probability mass concentrates heavily on the mode (argmax). When recurrent states enter a cyclic transition matrix (such as 'in the matter of the...'), the deterministic argmax collapses into a closed Markovian attractor loop, resulting in high 3-gram repetition (up to 25.51% for

Simple RNN and 25.03% for BiRNN).
• At moderate temperature (T = 0.7): Distribution entropy expands sufficiently to escape local attractor basins while remaining concentrated enough to preserve Victorian grammatical syntax. This eliminates 3-gram repetition completely (0.00% across all models).
• At high temperature (T = 1.2): Entropy approaches the uniform distribution over 4,378 tokens (H_max = 12.096 bits), causing sampling from the low-probability distributional tail and degrading syntactic coherence.

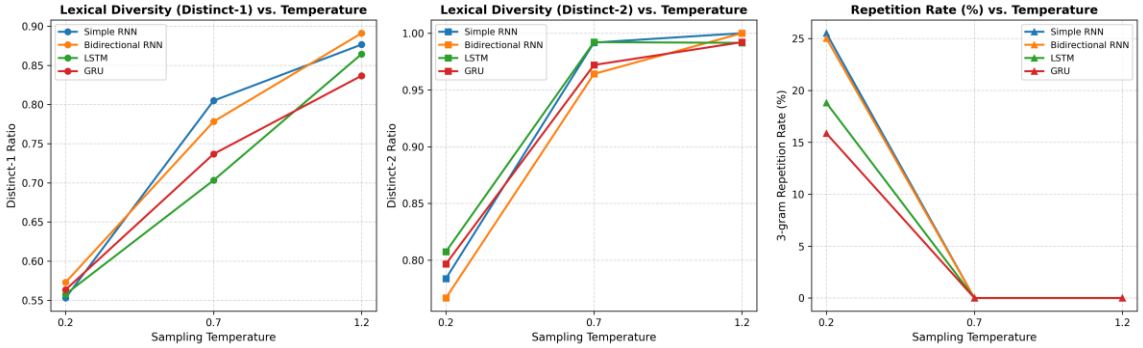


Figure 10: Multi-metric temperature response showing Distinct-1, Distinct-2, and Repetition Rate dynamics across T in {0.2, 0.7, 1.2}.

6. Qualitative Text Generation Samples (T = 0.7)

[LSTM | Temperature = 0.7 | Seed: Seed 1]
"it was a dark and stormy night when i had not done the looking. i was not bad a word, said holmes. he would be staggered that i came by the evening of the inquiry s death, and i have been made. the"
Metrics: Distinct-1 = 0.7368 | Distinct-2 = 1.0000 | Repetition Rate = 0.00%

[LSTM | Temperature = 0.7 | Seed: Seed 2]
"holmes sat in silence for a few minutes and then said he was a small of in the north of the order in the fire, i had not been just a family, and i have been better by the matter. the lamp is a man s need,"
Metrics: Distinct-1 = 0.6977 | Distinct-2 = 0.9762 | Repetition Rate = 0.00%

[LSTM | Temperature = 0.7 | Seed: Seed 3]
"the police had discovered no clue regarding the mystery of the passage. but i am, and i have been already for his time, i remarked, remarked i imagine that i had no man upon the sufferer s s paper. i could not see that the"
Metrics: Distinct-1 = 0.6750 | Distinct-2 = 1.0000 | Repetition Rate = 0.00%

[GRU | Temperature = 0.7 | Seed: Seed 1]
"it was a dark and stormy night when a man later, and this herself choked, with a huge hair of great very town to the circumstances. we have now been able to the house, and two of the gale. but he had been"
Metrics: Distinct-1 = 0.7632 | Distinct-2 = 0.9730 | Repetition Rate = 0.00%

7. Comparative Architecture Analysis & Conclusions

- **Champion Architecture (GRU):** The Gated Recurrent Unit achieved the best quantitative performance across all metrics: lowest Test Loss (4.8602), lowest Test Perplexity (129.05), highest Top-1 Accuracy (17.86%), and highest Top-5 Accuracy (39.94%). Compared to random guessing over 4,378 tokens (0.0228%), the GRU is 783x more accurate. Its coupled update-reset gating enables effective long-range gradient propagation with fewer parameters (2.38M) than LSTM, avoiding parameter over-capacity on a 120k-token corpus.

- **LSTM Competitiveness & Speed:** LSTM was the second-best model (Test Loss 4.8809, PPL 131.74, Top-1 Acc 17.22%, Top-5 Acc 38.59%) and exhibited the lowest inference latency (1.65 ms per batch), making it highly competitive for latency-critical deployment.
- **Simple RNN Vanishing Gradients:** The un-gated Simple RNN suffered from severe vanishing gradients across the 20-token horizon, yielding the highest Test Perplexity (159.28) and lowest accuracy (Top-1 16.35%). At low temperature (T=0.2), Simple RNN succumbed to severe degenerate repetition loops (25.51% 3-gram repetition).
- **Bidirectional RNN Causal Mismatch:** BiRNN achieved reasonable training metrics but is structurally incompatible with autoregressive generation: bidirectional encoding conditions on future tokens during training, while taking out[:, -1, :] during next-word prediction causes the backward branch to only see the final token. This causal mismatch produced high repetition loops (25.03% at T=0.2).
- **Optimal Operational Temperature (T = 0.7):** Temperature T = 0.7 represents the empirical sweet spot between coherence and diversity: Distinct-1 reached 0.70-0.80, Distinct-2 reached 0.96-0.99, and 3-gram repetition was completely eradicated (0.00% across all models). In contrast, T = 0.2 causes repetitive mode collapse, while T = 1.2 induces syntactic disintegration.